

LIF Neuron

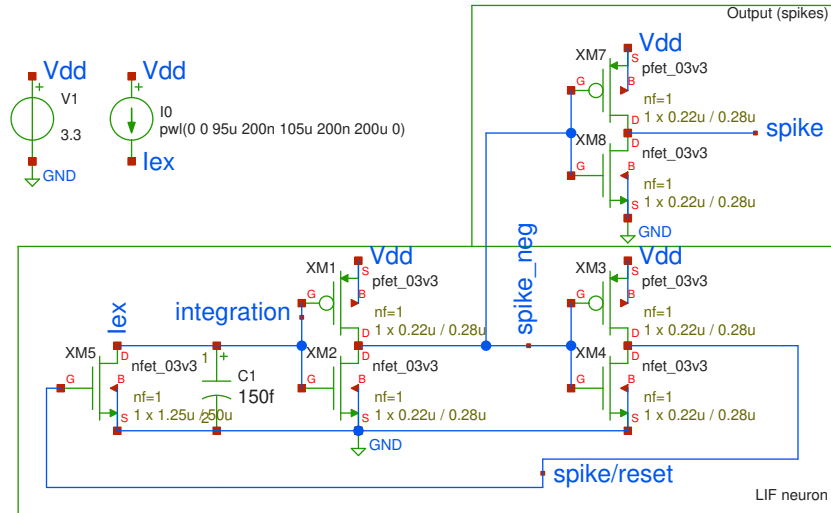
MODELS



```

spice
.tran 0.1n 200u
.control
save all
run
write neurona.raw
.endc

```



-The maximum frequency achieved is 1 MHz at 200 nA.

-The transistor M5 behaves like a resistor when it is off and leaks current when it is on.

-When the voltage in the transistor exceeds the threshold voltage (the inverter's switching voltage), a spike is generated.

-The output is obtained from the spike_neg node by inserting an additional inverter, so as not to disturb the neuron's spike/reset signal.

-A transient analysis was performed by varying the current source from 0 A to 200 nA and from 200 nA to 0 A.

